Evidence review: 15-position prime-norm scalar radix

Question. Does the 268−τ prime-norm radix provide a verified 15-position integer recoding, and what does the existing paired panel establish against the 16-position 190−τ format?

Scope and method. This report audits the frozen artifacts in this directory; it does not run a new benchmark. The search enumerated the declared T=240…320 prime-norm candidates, checked selector/radius/terminal bounds, compared the six-candidate selector with the bounded global search on all sign-canonical residues, and replayed the selected recoding. Integer checks are complemented by independent affine point replay and native full-residue/holdout tests. The performance comparison uses panel 2481bd3c0f7e491698694ccb039f1048 on RunPod instance thyvmdqmzsz5ik, comparing radix 15 (N=71,023) against radix 16 (N=35,533) on the same 128 frozen scalar inputs. No secp256k1 curve identity or subgroup claim is needed for the scalar-format comparison; the independent group replay is specifically over secp256k1.

Findings

Claim	Status	Evidence and limits
N(268−τ)=71,023 is prime and the exact declared sweep has twelve prime-norm candidates.	Derived and independently checked	frontier15.py, frontier15.json, NearestOrbit71023Bound.lean. The sweep establishes only the stated T=240…320 range.
The selected integer recoding has 15 steps and terminal norm below 82^2.	Proved for the formalized integer recoding	Selector, finite digit certificate, contraction, and trajectory declarations in the six Lean files linked by README.md; checked logs are retained. The candidate-specific secp256k1 group-action theorem remains unformalized.
The 15-position path matches independent point multiplication on the checked samples.	Independently verified on bounded samples	388 affine replays and a 128-scalar native holdout, documented in group-replay268.json, screen268.json, and the native test/build logs. This is sample-based group validation, not a proof of the curve-specific bridge for all inputs.

Claim	Status	Evidence and limits
Median sustained hot-table throughput was 1.024327× radix 16, with repeat-bootstrap 95% interval [1.020012, 1.031215].	Measured; the panel passes its A/A and interval gate, with a repository evidence requirement still missing	tier1-analysis.json, tier1-rules.json, tier1-manifest.json, and the 96 raw arms / 48 pair receipts under 2481bd3c0f7e491698694ccb039f1048/. The raw arm receipts omit per-process peak RSS, required by the base-branch hardware-evidence rules. The available cgroup peak is container-wide and not attributable to either process. The result is for repeated warm-table work on this allocation and frozen scalar law; it does not establish cold-start or end-to-end library speedup.
The candidate uses less setup or memory.	Refuted by recorded measurements	Mean preparation was 2,114.12 ms versus 1,257.82 ms and retained table memory was 12,336,756 versus 6,829,948 bytes (candidate/reference). These costs are outside the online timing interval and are shown separately.

The generic runner's archived summary.json is retained unchanged. Its paired_speedup value (1.003080) is the median from the pre-comparison A/A calibration case, and its generic speedup_claim_eligible=false describes that runner-level aggregate. The radix comparison result is separately derived by analyze_tier1.py from the compare case after checking both adjacent A/A calibration cases; the comparison median and interval are in tier1-analysis.json. The two summaries describe different calculations and must not be conflated.

The 96 raw arm receipts record child user/system CPU time, wall/online time, cgroup throttling and memory-event deltas, steal time, faults, and context switches. The audit found zero throttling, steal, major faults, or OOM events; involuntary context switches total 647 across all arms. The receipts do not retain the runner's per-child peak RSS field. The preflight records a 1,781,030,912-byte cgroup peak, but that is container-wide and cannot stand in for per-process peak RSS. Thus the frozen panel passes the comparison's service gate, while its retained data do not satisfy the repository's complete per-run memory-evidence requirement. Reconstructing that missing process statistic from the other counters is not valid.

The panel manifest records hashes for both benchmark binaries and source files. However, the candidate overlay requires a parent source snapshot whose hash (e195374c…) does not match the source file committed in the base tree; the manifest's external /workspace/bench-sources/… snapshots are not present in the run archive or this review workspace. The PR preserves the overlay procedure and hashes, but that historical native build cannot be reconstructed from the checked-out evidence alone. The input fixture itself is preserved in the base experiment tree and its SHA-256 matches the panel manifest.

Visuals and graph-set audit

figures.py is the editable source for the evidence-flow diagram and the quantitative throughput graph. Their SVG and PDF renders are alongside the source; REPORT-2026-10-19.pdf contains this

report and both visuals. The throughput figure uses the existing paired median and repeat-bootstrap interval and labels the sample, timing scope, tier, and host-isolation limit.

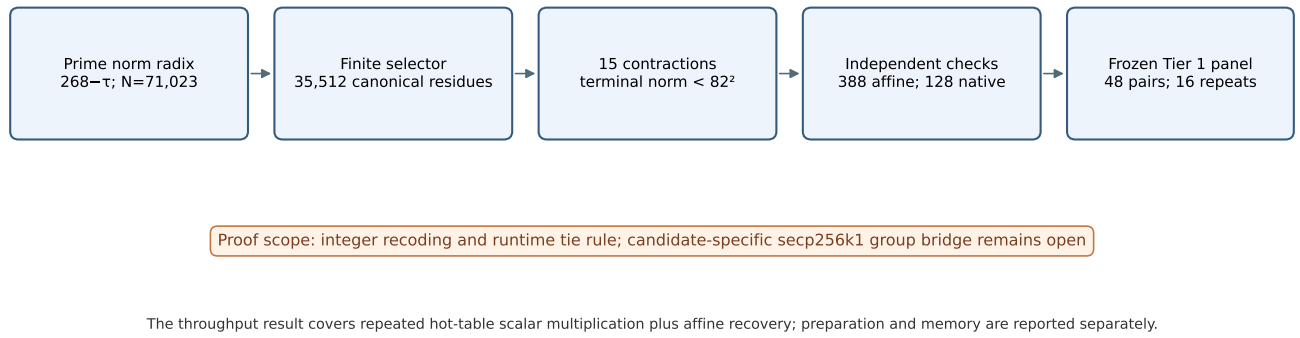


Figure 1: Evidence flow from the prime-norm radix through integer proof and bounded group checks to the paired panel.

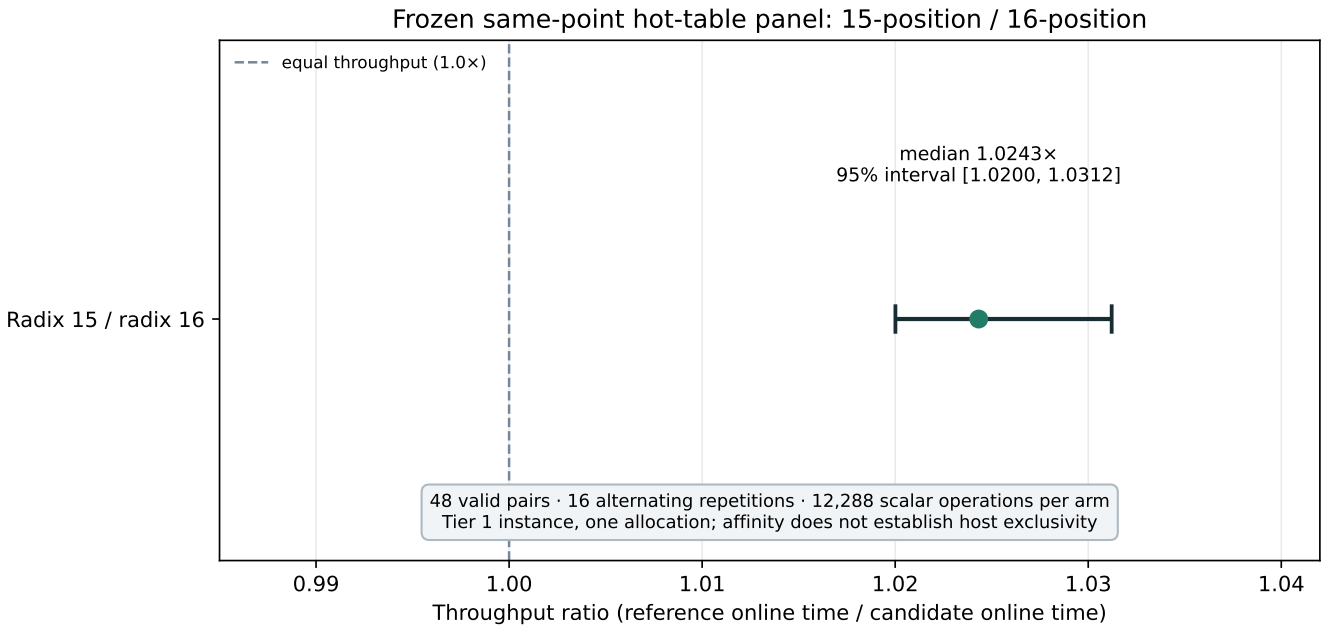


Figure 2: Measured hot-table throughput ratio and repeat-bootstrap interval for the 15-position candidate against the 16-position reference.

I checked the existing `experiments/ec-arithmetic-optimizations/` result figures and the `experiments/volcano-ic/figures/` graph set. The former compares arithmetic-layer workloads and has no matching radix-15/radix-16 observation to update; the latter tracks isogeny/IC surface structure, which this scalar-format result does not change. Neither canonical graph's source data or conclusion changes here. The local graph added above is the quantitative visualization for this newly measured, separate workload. No curve registry, IC route, benchmark baseline, or dashboard row is changed because this is not an IC candidate or production baseline measurement.

Limitations and open checks

The paired interval describes repetition variation on one cloud instance, not host-exclusive performance or variability across providers, CPUs, or scalar distributions. CPU affinity records process placement only. One-call latency was not measured. Table construction, preparation, and memory favor the 16-position reference. The formal proof covers the integer recoding; the curve-specific group bridge is supported by independent bounded replays but remains a formalization gap. These results do not establish production integration or end-to-end secp256k1 library speedup.

The exact source for the measured binaries is bound by the recorded source hashes and native overlay receipts; the raw panel archive contains the run receipts, while the PR preserves the overlay and source-generation scripts. Reproducing the historical remote allocation is not part of this review and no new performance run was performed.

Reproduction

Run `python3 figures.py` to regenerate the editable SVG/PDF visuals from `tier1-analysis.json` and `tier1-rules.json`. Run `python3 analyze_tier1.py` to regenerate the report's paired audit from the preserved panel receipts (the script imports the matching auditor from the 2026-10-14 parent experiment). To replay the integer and group checks, from the repository root set `PYTHONPATH=experiments/prime-j0-nearest-digit-12211-20261014:experiments/prime-j0-nearest-orbit-12211-20261014:experiments/prime-j0-mixed640-materialized-tree-20261013:experiments/prime-j0-mixed-radix640-20261010:experiments/prime-j0-tau-power16-20261010` before running `screen268.py` or `group_replay268.py`; outputs matched the frozen JSON values exactly in this review. The retained Lean logs show the six proof checks used for the declarations listed in `README.md`; Lean is unavailable in this review environment, and the candidate-specific point-group bridge is explicitly pending.